Exploring Uni-Modal Feature Learning on Entities and Relations for Remote Sensing Cross-Modal Text-Image Retrieval

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***Abstract*— Remote sensing cross-modal text-image retrieval (RSCTIR) has recently received unprecedented attention due to its advantages of flexible input and efficient query on enormous remote sensing (RS) images. However, most RSCTIR methods focus obsessively on the cross-modal semantic alignment between text and image modalities and are easily stuck in the dilemma of information redundancy, leading to the degradation of retrieval accuracy. To address the issues, we construct a novel RSCTIR framework based on mask-guided relation modeling with entity loss (MGRM-EL), to fully explore uni-modal feature learning on entities and relations in the cross-modal model learning process. Specifically, we take advantage of the Transformer encoder architecture for its ability to capture long-distance dependencies from the global view and build two uni-modal (visual and textual) Transformer encoders combined with convolutional neural network (CNN) to extract the spatial interregion relations of images as well as long-term interword relations of texts for prominent feature embedding of visual and semantic representations. A mask-guided attention strategy is further introduced to learn the salient regions and words, with the aim of enhancing the RSCTIR model's uni-modal learning ability and eliminating unnecessary and redundant information about each modality. Unlike existing methods that simply compute the semantic similarity between images and texts on the loss functions, we present a novel uni-modal entity loss, which treats each image as an image entity and merges similar texts into a text entity, to learn the independent distribution of entities in each modality. We conduct extensive experiments on public RSCTIR benchmarks including RSICD and RSITMD datasets, which demonstrate the state-of-the-art performance of the proposed method on the RSCTIR task.**

***Index Terms*— Mask-guided attention strategy, relation exploration, remote sensing cross-modal text-image retrieval (RSCTIR), uni-modal entity loss.**

I. Introduction

REMOTE sensing (RS) is the science and technology of acquiring and recording information about physical objects, areas, or phenomena from a distance, typically from aircraft or satellites [1]. In recent years, the rapid developments of RS applications in the fields of agriculture, forestry, environment, and urban planning, have led to an explosive growth of RS image data [2]. However, in contrast to the RS image acquisition ability, the RS data processing capability is very weak and easily falls into the paradox of "big data, little knowledge" [3], [4]. To obtain valuable knowledge from enormous RS data in a convenient and flexible way, effective automatic RS cross-modal retrieval methods using different modal information are increasingly needed [5]. Due to its advantages of fast extraction and filtration of image information and flexible human–computer interaction, RS cross-modal text-image retrieval (RSCTIR) [6], [7] provides a possibility to explore the growing amount of cross-modal RS data, and becomes an emerging research area at the intersection of natural language processing (NLP) and computer vision (CV).

The RSCTIR task contains two subtasks, which are image-to-text (I2T) retrieval and text-to-image (T2I) retrieval [8]. The I2T subtask is defined as an image query to return a caption description that matches its visual content, while the T2I subtask is vice versa. RSCTIR aims at efficiently retrieving objects of interest from multimodal RS data and enables computers to understand image-text pairs in semantic terms [2]. However, the huge semantic gap in feature representations between the modalities of images and texts makes it significantly challenging. Therefore, various RSCTIR methods based on deep learning have been proposed to establish connections between image and text modalities. For example, Abdullah et al. [6] proposed a deep bidirectional triplet network to learn the joint encoding between multiple modalities. Yuan et al. [9] constructed an asymmetric multimodal feature matching network (AMFMN) to solve the problem of multiscale scarcity and target redundancy. Yuan et al. [10] presented a framework based on global and local information (GaLR), and designed a multilevel information dynamic fusion (MIDF) module. Although these methods have recently made progress in the development of RSCTIR, there are still three issues overlooked, but inevitable, when these methods are applied to RSCTIR.

First, RS images often contain a large number of irrelevant background regions that are unrelated to objects of interest, while there are also some redundant words in the text sequence. This information redundancy leads to the distraction on the matching of images and texts and prevents the retrieval model from understanding the content of RS images and texts. In this article, with the aim of eliminating unnecessary and

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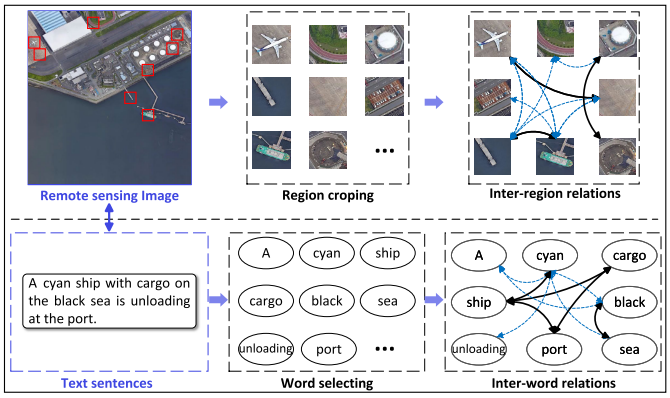


Fig. 1. Visual learning for the spatial relation of image regions and the semantic relation of text words in the RSCTIR task. The black solid line represents the strong potential intramodal relation, and the blue dashed line denotes the weak relation.

redundant information and enhancing the model's uni-model learning ability, we present a mask-guided attention strategy, which randomly samples a small set of patches from image regions or randomly replaces some input words with a special word (named [Mask]) in the training stage, to extract salient feature expressions from images and texts.

Second, most current methods ignore the two intra-modal inherent semantic relations: the spatial interregion relations in the image modality and the long-term interword relations in the text modality. As shown in Fig. 1, some regions are spatially related and have implicit intra-modal relations in the image, while some words are semantically associated with the intramodal relations in the text. For example, in the RS image, the "ship" region is more closely related to the one of "harbor" than to the "parking lot." In the text sequence, the relationship between "ship" and "port" is closer than that between "cyan" and "sea." In this article, to explore intramodal relations for each modality, we present a uni-modal relation modeling method that builds two uni-modal (visual and textual) Transformer encoders [11] to extract the spatial semantic relations of images and long-term word dependencies of texts for prominent feature embedding of visual and semantic representations from the global view.

Third, unlike natural scenes in which multiple captions corresponding to one image are highly relevant for describing the image content, remote-sensing images obtain numerous objects and multiple captions may indicate entirely different patches of the RS image. As shown in Fig. 2(a) in the natural scenes dataset of Flickr30k [12], each image has five relevant captions that describe the salient objects in the image, while in Fig. 2(b) in the RS scenes dataset of AIR-SLT [13], the five captions corresponding to one RS image point to five different areas, respectively. However, existing RSCTIR methods [6], [9], [10], [14] often treat the five captions corresponding to one RS image as one entity in the same way as that in natural scenes, and only compute the cross-modal similarity as shown in Fig. 2(c). Consequently, those methods will be ambiguous about the optimization objective during training. In order to alleviate this issue, we propose a novel uni-modal entity loss, which treats each image as an image entity and merges similar texts into a text entity with the text clustering algorithm. The uni-modal entity loss employs cross-entropy classification loss to enforce the model to discriminate image and text entities so as to learn the independent distribution of entities in each modality, as shown in Fig. 2(d).

The main contributions of our work are summarized as follows.

1) Unlike existing RSCTIR methods that focus on cross-modal semantic alignment, we fully explore uni-modal feature learning on entities and relations, and construct a novel end-to-end framework based on mask-guided relation modeling with entity loss (MGRM-EL).
2) Two uni-modal (visual and textual) Transformer encoders combined with the mask-guided self-attention strategy are introduced to better explore spatial relations of image regions and semantic relations of text words.
3) We present a novel uni-modal entity loss to optimize the training objective by simultaneously learning the semantic alignment between images and texts and discriminating interentity independence discrepancy.

MGRM-EL achieves the new state-of-the-art performance on two popular datasets (RSICD and RSITMD), as shown in Fig. 3, which demonstrates the capability of the proposed model on uni-modal relation modeling and cross-modal image-text matching.

The rest of this article is organized as follows. In Section II, we briefly summarize previous related works in RSCTIR. In Section III, we introduce the implementation details of our proposed MGRM-EL and the uni-modal entity loss. Furthermore, we give a fair comparison between the proposed model and other baseline models in Section IV. Finally, we conclude the work in Section V.

II. Related Works

In this section, we briefly discuss recent works on RSCTIR, Transformer-based methods, masked language/image modeling, and loss functions for RSCTIR.

A. RS Cross-Modal Text-Image Retrieval

RSCTIR is an active research hotspot that aims to retrieve relevant visual and textual information from the RS data. In recent years, there has been a growing interest in this field, and various methods have been proposed. In terms of implementation methods, RSCTIR tasks can be generally divided into caption-based methods and embedded-based methods.

The caption-based methods tend to use a caption generator to generate text labels from each RS image and obtain the retrieval results by calculating the similarity score between the query text and the generated labels in the text modality with the bilingual evaluation understudy (BLEU) [15] metric. Qu et al. [16] first utilized multimodal deep networks for the semantic understanding of high-resolution RS images. Shi and Zou [17] proposed an RS image captioning (RSIC) framework based on deep learning to semantically decompose ground elements at different scales. Lu et al. [18] proposed a large RSIC dataset and conducted a comprehensive review

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Fig. 2. Comparison of cross-modal image-text retrieval in the natural scene and RS scene. (a) Natural images have five relevant captions that describe the same salient objects and events. (b) Captions of RS images indicate entirely different areas since RS images contain more information. (c) Traditional RSCTIR methods generally use the triplet loss that only calculates the cross-modal distance. (d) We present a uni-modal entity loss to learn the independent distribution of entities in each modality. The symbols ■ and ● with the same color denote the matched image-text pairs. The boxes and circles with the green color denote the image and texts from (a), while the boxes and circles with the blue color mean the image and texts from (b).

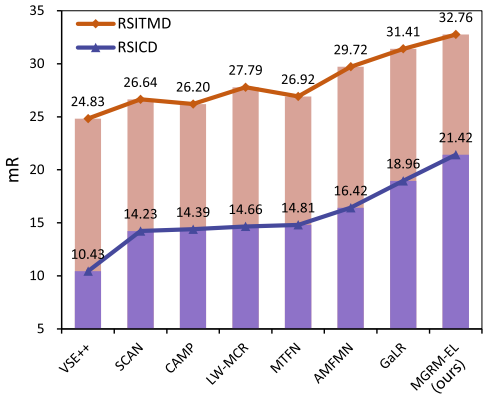


Fig. 3. Comparison of the proposed MGRM-EL with other state-of-the-art (SoTA) RSCTIR methods on the RSITMD and RSICD datasets. The proposed method achieves the mR metric of 32.76 on RSITMD and 21.42 on RSICD, respectively, significantly outperforming the SoTA methods.

to advance the task of RSIC. Despite the relative maturity of caption-based methods [19], [20], [21], these methods inevitably result in the information loss due to the cross-modal transformation [10].

The embedded-based methods measure the cross-modal similarity by calculating the appropriate feature distance of images and texts in a single stage. Abdullah et al. [6] proposed a deep bidirectional triplet network to learn joint encodings among multiple modalities. Cheng et al. [7] proposed a semantic alignment module to get a more discriminative feature representation and explore the correspondence between RS images and texts. Lv et al. [22] suggested a fusion-based correlation learning model for RS image-text retrieval, and solved the heterogeneity gap problem by knowledge distillation. To cope with the problem of multiscale scarcity and target redundancy, Yuan et al. [9] came up with a novel AMFMN. To handle the issue of current methods that neglect local features, Yuan et al. [10] proposed a method based on GaLR and designed a MIDF module to enhance visual representations.

Although the above-mentioned methods have succeeded in the development of RSCTIR, they generally focus on bridging the cross-modal semantic gap between text and image modalities. Differently, the proposed method aims at exploring the uni-modal feature learning on entities and relations during the RSCTIR model learning process by extracting the spatial relations in image regions and the semantic relations in text sequences with two uni-modal Transformer encoders and learning the independent distribution of entities in each modality with the proposed entity loss.

B. Transformer-Based Methods

1) Transformer-Based Methods in NLP and CV: Transformer-based methods [23] were originally widely used in NLP [24], and have recently demonstrated excellent performance on CV tasks [11]. The success of Transformers can be mainly attributed to self-attention [25], which takes the context of a given sequence into account by learning the relations between the elements in the token set (e.g., words in a sentence or patches in an image). BERT [24] is one typical Transformer-based method in NLP, which pretrains on unlabeled text data and fine-tunes downstream tasks. As a notable Transformer-based method in CV, the vision transformer (ViT) [11] splits

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an image into multiple patches and feeds them into a stacked Transformer architecture for classification. The development in NLP and CV demonstrates the power of Transformer-based models in achieving state-of-the-art performance and capturing complex relationships in their respective domains.

2) Transformer-Based Methods in RS: Inspired by recent advances in CV, the RS community has also witnessed an increased exploration of ViTs for a diverse set of tasks [26], [27], [28], such as very high-resolution image classification, change detection, pan sharpening, building detection, and image captioning. For example, CCTNet [29] combines local details such as edges and textures captured by convolutional neural network (CNN) with global contextual information obtained through a transformer for segmentation in RS images. Liu et al. [30] introduced an encoder-decoder architecture with transformers for RSIC, and a multilayer aggregated transformer is utilized in the decoder to effectively exploit the multiscale features for generating sentences.

3) Transformer-Based Methods in RSCTIR: Some recent methods attempt to apply Transformer-based methods to solve the challenges in RSCTIR. Zhang et al. [31] proposed a transformer-based cross-modal retrieval of RS images method using feature decoupling and reconstruction (TBFDR) to obtain the modal-invariant features. However, TBFDR only uses Transformer as a feature extractor. GaLR [10] also utilizes the capability of self-attention in Transformer, but it applies Transformer to enhance only image features. Al Rahhal et al. [32] proposed a multilanguage framework based on Transformer that used multilingual queries to enhance communication with the retrieval system and broaden access to the RS information. Tang et al. [33] proposed a model named interacting-enhancing feature transformer (IEFT) that leveraged Transformer to deeply model the relationship between images and text across modalities. Recently, Rahhal et al. [34] proposed a text-image retrieval approach based on vision and language transformers. These above methods applied the transformer architecture to extract global feature learning, however, they generally took transformer as feature extractor and focused obsessively on the cross-modal semantic alignment between text and image modalities. In this article, our method MGRM-EL explicitly explores uni-modal feature learning on both interregion relations of images and interword relations of texts and thus leverages two uni-modal Transformer encoders (visual and textual) to learn prominent feature embeddings. Based on this, we apply a mask-guided Transformer method to eliminate redundant information in RS data. Furthermore, we exploit the self-supervised classification signal and present an entity loss to learn the deep network features of different identities for each modality.

C. Masked Language/Image Modeling

Masked language modeling was originally proposed in the field of NLP, such as some highly successful methods like BERT [35], and chatGPT [36], [37], [38] which has generated an explosion of interest. Some researchers in CV have also proposed the masked language modeling mechanism. BEiT [39] based on ViT [11] proposed to predict discrete tokens. He et al. [40] proposed masked autoencoders (MAEs), which masked random input image patches and could reconstruct the missing pixels. Most recently, Bachmann et al. [41] proposed a pretraining strategy called multimodal multitask MAEs (MultiMAEs), which randomly selected 1/6 of all 16×16 image patches from multiple modalities and learned to reconstruct the remaining 5/6 masked patches from them. Ren et al. [42] designed a mask-cross-entropy strategy in RSIC, which randomly replaced some input words with a special word in the training stage. For the RSCTIR task, there are few works that applied masked language/image modeling. In this article, according to the characteristics of RS data, we propose to utilize masked-guided attention to enhance the RSCTIR model's uni-modal learning ability and eliminate unnecessary and redundant information about each modality.

D. Loss Functions for RSCTIR

With the development of multimodal feature alignment, triplet loss has emerged as the mainstream loss function for the RSCTIR task [10]. It aims to minimize the distance between text-image paired samples while maximizing the distance between unpaired samples within a batch during training [43]. By enforcing a fixed margin, triplet loss encourages a compact embedding space where similar samples are brought closer together and dissimilar samples are pushed farther apart. Yuan et al. [9] designed a triplet loss function with a dynamic variable margin based on the sample pairs' prior similarity to alleviate the positive sample ambiguity caused by the strong intraclass similarity in the RS image. However, the commonly used triplet loss only calculates the cross-modal distance and considers a weak label (similar or not) between pairs, which is difficult to handle with hard samples. In this article, a novel uni-modal entity loss is proposed to learn semantic differences within a single modality. We treat each image as an entity to learn to discriminate image features of different identities. Meanwhile, considering that there may be similar or repeated content in the five text descriptions corresponding to an image in the retrieval task, we use a clustering algorithm to merge similar or repeated texts into an entity and discriminate features of texts with different semantics.

III. Proposed Method

In this section, we first describe the overall framework of the proposed MGRM-EL. Then we introduce the formulation of the feature extraction, the logic of the proposed mask-guided relation Transformer and the proposed uni-modal entity loss in detail.

A. Overall Framework

Fig. 4 illustrates the architecture of our proposed MGRM-EL for RSCTIR. Our method aims at fully exploring uni-modal feature learning on entities and relations in the cross-modal model learning process. We introduce our method in three main parts.

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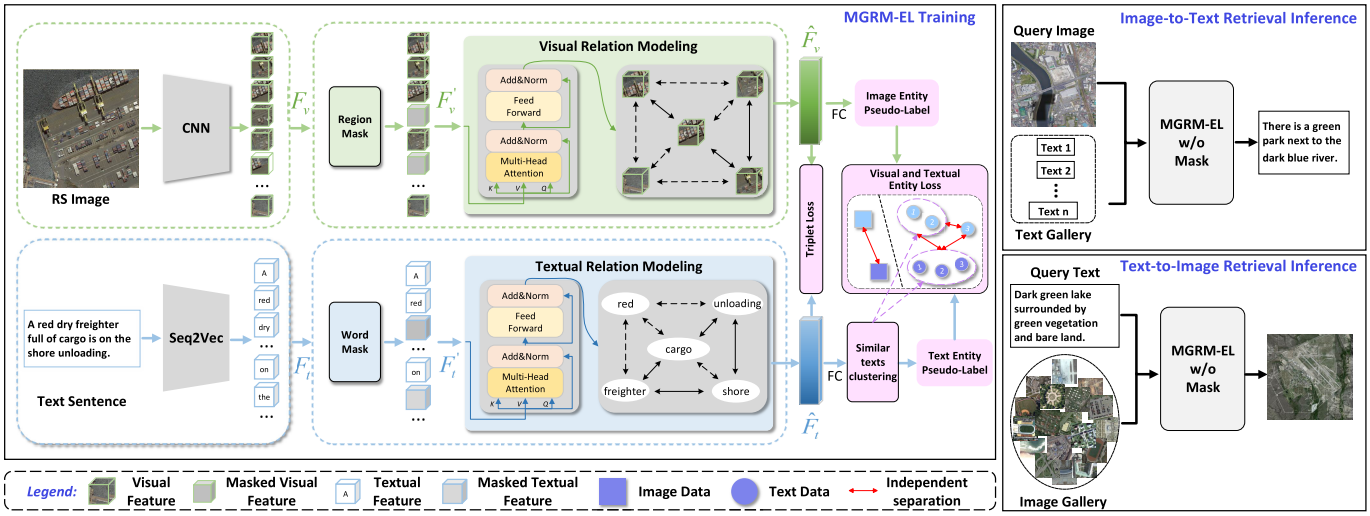


Fig. 4. Overall framework of the proposed MGRM-EL. (Left) MGRM-EL training: with the input image and text, the framework is constructed with three steps: extract visual and textual features, mask-guided relational modeling, and loss calculation. The mask-guided relational modeling process includes mask-guided feature learning and uni-modal relation modeling based on transformer encoders. A novel uni-modal entity loss is presented to learn the independent distribution of entities in each modality. (Right) Retrieval inference: RSCTIR can be divided into two subtasks, I2T and T2I. Note that during the inference process, all features will not be masked.

1) Visual and Textual Feature Extraction: We leverage a CNN backbone (e.g., ResNet18 [44]) to extract visual features F_v from input images I. For the input text sentences T, we utilize Skipthought [45] in the field of NLP as the textual feature extractor and obtain features F_t.

2) Uni-Modal Mask-Guided Relational Modeling: The modeling process is divided into two steps, including mask-guided feature learning and uni-modal relation modeling. Mask-guided feature learning randomly samples a small set of patches from image regions or some input words from text words, to enhance the feature learning ability and eliminate unnecessary and redundant information. Uni-modal relation modeling applies the capability of the Transformer's self-supervision and self-attention to model the intraregion and intraword relations, respectively. We finally obtain visual features $\hat{F}_v$ and text features $\hat{F}_t$ that contain prominent semantic information.

3) Uni-Modal Entity Loss: Unlike existing methods that apply the traditional triplet loss by only computing the cross-modal similarity between images and texts, we pay more attention to the independent distribution of entities in each modality, i.e., each image should be a unique image entity and the similar texts are merged into a text entity. We present a novel uni-modal entity loss combined with the triplet loss to optimize the training objective by simultaneously learning the semantic alignment between images and texts and discriminating interentity independence discrepancy.

B. Visual and Textual Feature Extraction

1) Visual Feature Extraction: To extract visual feature information from RS images, we apply the CNN backbone, ResNet18 [44] pretrained on ImageNet [46], as the visual feature extractor. Given one input RS image $I \in \mathbb{R}^{H\times W\times 3}$, the feature extraction process can be formulated as follows:

$$F_v = \mathcal{B}(I; \theta_v) \tag{1}$$

where $\mathcal{B}$ is the CNN backbone, and θ_v represents the learnable parameters of the CNN backbone. The output feature embedding is represented as $F_v \in \mathbb{R}^{h\times w\times d}$, in which h, w, and d are the height, width, and channels, respectively. Then we consider one feature vector $v_i \in \mathbb{R}^d$ from F_v, with i ranging from 1 to N (where $N = h \times w$), i.e., $F_v = \{v_1, v_2, \ldots, v_N\}$, $F_v \in \mathbb{R}^{N\times d}$.

2) Textual Feature Extraction: We apply Skip-thought [45], a Seq2Vec [47] model for unsupervised sentence representation learning, as the backbone to extract textual feature information from RS texts. Assume that one text description T contains a total of M words, i.e., $T = \{w_1, w_2, \ldots, w_M\}$. We use GRU [48] as an encoder and have the final text representation

$$F_t = \mathrm{GRU}(\mathrm{Emb}(w_1), \mathrm{Emb}(w_2), \ldots, \mathrm{Emb}(w_M)) \tag{2}$$

where $\mathrm{Emb}(w_i)$ represents the embedding vector of the ith word w. In this way, we can get the textual feature $F_t = \{t_1, t_2, \ldots, t_M\}$, $F_t \in \mathbb{R}^{M\times d}$.

C. Uni-Modal Mask-Guided Relation Modeling

With the visual features F_v and textual features F_t as input, uni-modal mask-guided relation modeling aims to explore more appropriate relations and derive robust feature representations for each modality. During the modeling process, a mask-guided attention strategy is first utilized to alleviate the redundant information in RS images and texts, and then we leverage the self-attention mechanism of the Transformer encoder [11] to model the long-range dependencies of the interregion relations of images as well as the interword relations of texts.

1) Mask-Guided Attention Strategy: For each modality, we randomly sample a subset of feature vectors and mask (i.e., remove) the remaining ones, following a uniform distribution. Random sampling with a masking ratio τ (i.e., the ratio of removed vectors) largely eliminates unnecessary and

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redundant information. With the visual feature F_v as inputs, we get a tensor $F_v' = \{v_1', v_2', \ldots, v_n'\}$, $v_i' \in \mathbb{R}^d$ as the feature embedding to represent sub-regions mapped back to original image space, where $n = \text{round}((1-\tau_v) \times N)$. For the textual feature F_t, we take a tensor $F_t' = \{t_1', t_2', \ldots, t_m'\}$, $t_i' \in \mathbb{R}^d$ as the feature embedding to represent words mapped back to original text sentence, where $m = \text{round}((1-\tau_t) \times M$.

2) Interregion Relations of Images: To model interregion relations of the masked feature embeddings F_v', we introduce a visual Transformer encoder to compute a normalized scalar attention coefficient α_{ij} between the ith and jth regional features, v_i' and v_j'. Then each feature vector v_i' is updated to $\hat{v}_i$ by calculating the weighted sum of all embeddings and following a nonlinear ReLU layer:

$$\alpha_{ij} = \text{softmax}\left(\left(W_r^Q v_i'\right)^{\mathsf{T}}\left(W_r^K v_j'\right)/\sqrt{d}\right) \tag{3}$$

$$\hat{v}_i = \text{ReLU}\left(\left(\sum_{j=1}^{n} \alpha_{ij} W_r^V v_j'\right) + b_1\right) + b_2 \tag{4}$$

where W_r^Q is the query weight matrix, W_r^K is the key weight matrix, W_r^V is the value weight matrix, and b_1 and b_2 are bias vectors. This update procedure can be repeated for x layers and the updated embedding $\hat{v}_i$ is fed as inputs to the successive Transformer encoder layer. The learned weight matrices $\{W_r^Q, W_r^K, W_r^V\} \in \mathbb{R}^{d\times d}$ are not shared between layers. We denote the final output of the Transformer encoder as $\hat{F}_v = \{\hat{v}_1, \hat{v}_2, \ldots, \hat{v}_n\}$, $\hat{v}_i \in \mathbb{R}^d$. With the multihead self-attention (MSA), the relational modeling of the above-mentioned regional features can be represented as

$$\hat{F}_v = \text{MSA}_{r\to r}\left(\left(v_i', \{v_1', \ldots, v_n'\}\right); \left(W_r^Q, W_r^K, W_r^V\right)\right) \tag{5}$$

where $\text{MSA}_{r\to r}$ represents the calculated multihead attentions between region features.

3) Interwords Relations of Texts: To model interwords relations of texts, we feed the masked textual feature embeddings F_t' into the textual Transformer encoder to learn relations between words, and get the output textual feature $\hat{F}_t$

$$\hat{F}_t = \text{MSA}_{w\to w}\left(\left(t_i', \{t_1', \ldots, t_m'\}\right); \left(W_w^Q, W_w^K, W_w^V\right)\right) \tag{6}$$

where $\text{MSA}_{w\to w}$ represents the calculated multihead attention between word features, and W_w^Q, W_w^K, and W_w^V represent the learned weight matrices of query, key, and value about words, respectively.

D. Uni-Modal Entity Loss

To fuse the multimodal representations for extracting the sentiment implications produced by multiple modalities, existing RSCTIR methods [6], [9], [10], [22] generally leverage a common triplet loss to transform the multimodal features into a shared feature representation space, so that the distance of the matched image-text pairs are pulled closer while that of the unmatched pairs are pushed away. However, these methods often ignore the independent distribution of entities in each modality. In this section, we first introduce the common triplet loss used in the RSCTIR task and its limitations. Then we describe the motivation, interpretation, and formulation of the proposed uni-modal entity loss.

1) Triplet Loss: Triplet loss is a common loss function used to learn similarity measures. For the RSCTIR task, triplet loss can be used to learn a shared vector space, in which image and text representations are mapped. The formula is as follows:

$$L_{\text{triplet}} = \underbrace{\sum_{\bar{F}_t}\left[\alpha - \cos(\hat{F}_v, \hat{F}_t) + \cos(\hat{F}_v, \bar{F}_t)\right]_+}_{\text{visual}} + \underbrace{\sum_{\bar{F}_v}\left[\alpha - \cos(\hat{F}_v, \hat{F}_t) + \cos(\bar{F}_v, \hat{F}_t)\right]_+}_{\text{textual}} \tag{7}$$

where α represents the minimum margin, and $[x]_+ = \max(x, 0)$. It achieves similarity optimization by shortening the distance between similar samples $\hat{F}_v$ and $\hat{F}_t$ and pushing the distance between heterogeneous samples $\bar{F}_t$ and $\bar{F}_v$. For the image-text positive pair, $\hat{F}_v$ represents the visual feature and $\hat{F}_t$ represents the textual feature. For the negative pair, $\bar{F}_t$ is the textual feature not paired with the visual feature $\hat{F}_v$, while $\bar{F}_v$ is the visual feature not paired with the textual feature $\hat{F}_t$. The cosine function $\cos(\hat{F}_v, \hat{F}_t)$ represents the similarity of visual features and textual features.

2) Limitation: Although the triplet loss in (7) is widely used in the RSCTIR task and achieves some performance improvement, note that it only computes the cross-modal distance between the two modalities and considers a weak label (similar or not) between pairs, which leads to the degradation of retrieval accuracy especially on some hard samples. In our experiments (as shown in Section IV-C), we also observe that for those test images or sentences with slightly different semantics, RSCTIR models are easily confused.

3) Motivation: The CNN-based approaches [49], [50], [51] in the field of CV, such as image classification, face recognition, and person reidentification, are mainly based on the identification mode. The identification mode makes use of the label information of the dataset to learn a mapping function from raw images to high-level identities/classes directly. Recently, some methods [52], [53], [54], [55] apply the multitask learning mechanism by jointly using the classification supervision and the verification supervision in triplets and emphasize different aspects of feature learning: the classification supervision requires features of different identities to be distinct, while the verification supervision pushes features from the same category closer and pulls apart those from different categories. Therefore, we consider improving the RSCTIR loss function which can explicitly learn the deep network features of different identities in a single modality.

4) Interpretation: There have been few works that apply classification loss in the RSCTIR task, since the images and texts in the RSCTIR dataset lack category-level annotations for supervised learning. To alleviate this issue, we attempt to exploit the self-supervised classification signal and then optimize the training objective by learning the deep network features of different identities for each modality.

For the image modality, each RS image is obviously different and thus each RS image can be treated as an image entity with an independent image class identity. With image classification supervision, we can employ the classification

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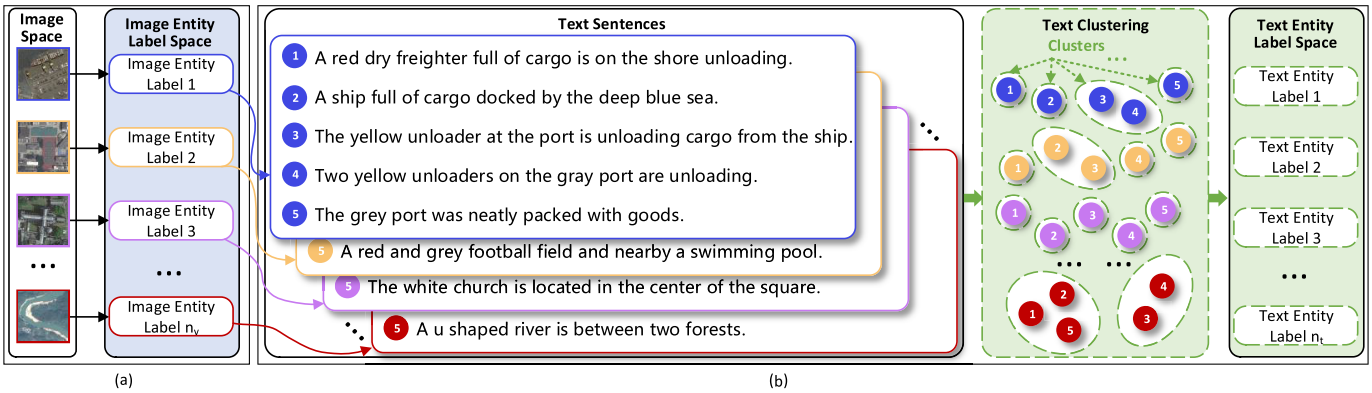


Fig. 5. Pseudo-label generation for uni-modal entity loss. (a) One-to-one image entity pseudo-label generation. Due to the independence of RS images, we treat each RS image as an entity. (b) Text entity pseudo-label generation. There may be similar or duplicate content in the text descriptions corresponding to the same image. We use the text clustering algorithm to generate pseudo-labels and treat each cluster class as a text entity.

loss to learn discriminative visual feature representations. For the text modality, an alternative method is that each sentence is treated as an independent text entity and all sentences are assigned with different labels. However, the fact is that there may be similar or duplicate contents in the text descriptions corresponding to the same image in the RSCTIR dataset. Consequently, to exploit more accurate pseudo-labels for textual features, we merge similar texts into one entity with a text clustering algorithm to aggregate the texts with the same content into clusters and assign each cluster with one pseudotextual label. As shown in Fig. 5, we generate pseudo-labels for image and text modalities, respectively. Therefore, the RSCTIR model with the proposed uni-modal entity loss can be trained with the generated pseudo-labels to distinguish image and text entities, respectively, by classifying entities of images/texts into different classification identities.

5) Formulation: With the obtained visual feature $\hat{F}_v$ and textual feature $\hat{F}_t$, we first classify them through the fully connected layer and then calculate the cross-entropy loss with pseudo labels. For images, since each image is different, we treat each image as an entity to generate the pseudo-label $l_v = \{1, 2, \ldots, n_v\}$, where n_v represents the number of pseudo-labels of images and it is equal to the number of images. For texts, there may be repeated or similar texts among the five texts corresponding to an image. We cluster similar or repeated texts into one category via the feature clustering method and treat them as an entity. Specifically, we apply the clustering algorithm of density-based spatial clustering of applications with noise (DBSCAN) [56] to aggregate the texts with the same content into clusters. DBSCAN is an unsupervised density clustering algorithm, which serves as a potential alternative to popular clustering techniques like KMeans [57] and hierarchical clustering [58]. It aims at identifying data points with a significantly high density and then group them into clusters. DBSCAN estimates the data density through two important parameters including the radius (ε) and the minimum number of data points (minPts). The pseudo-text-label generation can be formulated as

$$l_t = \text{DBSCAN}(T;\ \varepsilon,\ \text{minPts}) \tag{8}$$

where l_t represents the final generated text pseudo-label $l_t = \{1, 2, \ldots, n_t\}$, n_t represents the number of categories of the clustered text, and T represents the input text features. The impact of parameter settings on model performance is also discussed in Section IV-H.

For the visual branch, using one-hot encoding of the labels l_v, we formulate the classification objective as follows:

$$P_{\text{visual}} = \text{softmax}(\mathbb{W}_v \hat{F}_v + b_1) \tag{9}$$

$$L_{\text{visual}} = -\sum_{i=1}^{n_v} l_v^i \log\left(P_{\text{visual}}^i\right) \tag{10}$$

where $\mathbb{W}_v$ and b_1 are weights and bias of the fully connected layer, l_v is one-hot pseudo-labels for images, P_{visual} is the predicted possibility of every class and L_{visual} denotes the loss of the visual branch. Similarly, we have L_{textual} that computes the loss of the textual branch

$$P_{\text{textual}} = \text{softmax}(\mathbb{W}_t \hat{F}_t + b_2) \tag{11}$$

$$L_{\text{textual}} = -\sum_{i=1}^{n_t} l_t^i \log\left(P_{\text{textual}}^i\right). \tag{12}$$

The final Entity loss can be defined as

$$L_{\text{Entity}} = L_{\text{visual}} + L_{\text{textual}}. \tag{13}$$

The final loss function of MGRM-EL is a combination of the triplet loss and the uni-modal entity loss, defined as

$$L = L_{\text{triplet}} + \lambda L_{\text{Entity}} \tag{14}$$

where λ is the predefined weight parameter for the entity loss. Specifically, λ is set to 1.0 in the experiments. In Section IV-H, we evaluate the influence of different values of λ on the image-text retrieval performance of the model.

E. Training and Inference Procedure

The training and inference procedures of the proposed MGRM-EL are shown in Algorithms 1 and 2, respectively. The comprehensive explanation of the algorithm is given as follows.

In the training stage, for each image-text pair $(I, T) \in D_{\text{train}}$ of the training set, we first apply ResNet18 and Skip-thought as the backbone to extract visual features F_v and textual features F_t, respectively. Then the mask-guided attention strategy

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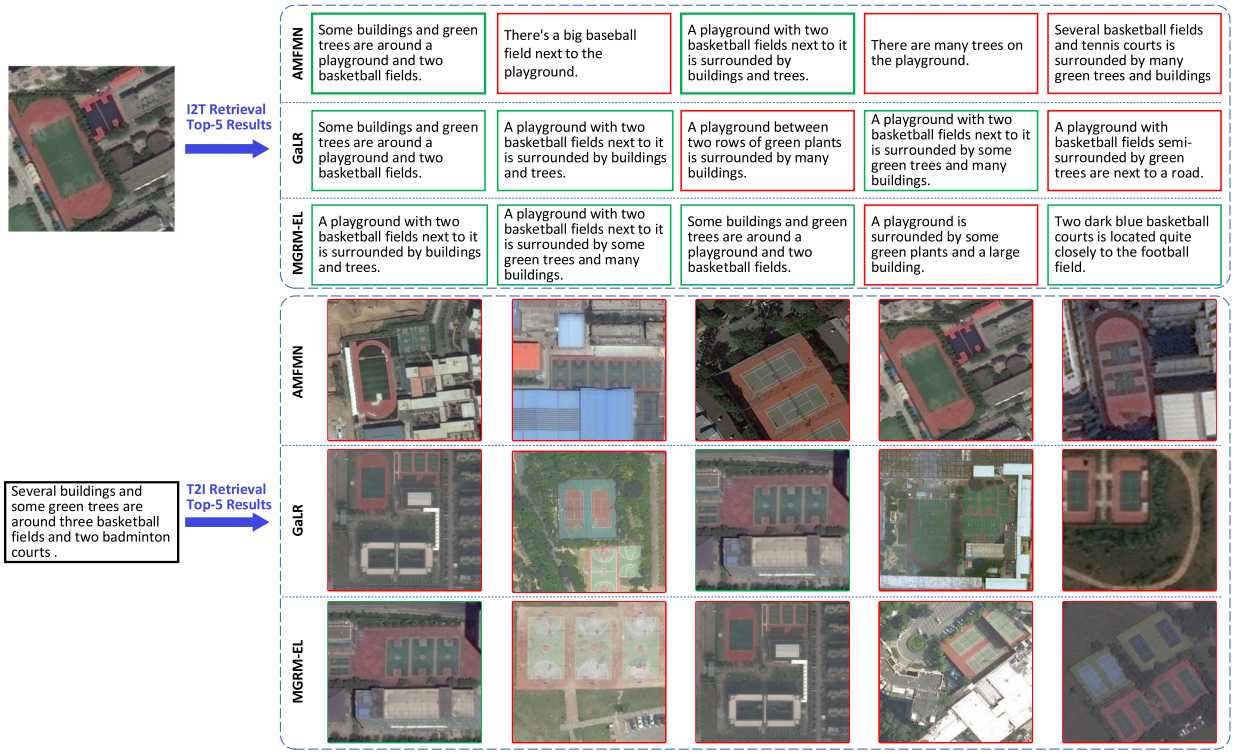


Fig. 6. Visualization of retrieval results on the RSICD testset. We, respectively, show the retrieval results of each model when performing I2T and T2I retrieval tasks. In the top 5 results, those surrounded by green boxes are ground truth, and those surrounded by red boxes are wrong results. It should be noted that there are five correct retrieval texts for each image, but only one correct retrieval image for a text.

is utilized to mask the input visual and textual features to learn the salient regions and words with the outputs F_v' and F_t'. After that, two uni-modal (visual and textual) Transformer encoders with the MSA module are leveraged to extract the spatial interregion relations of images as well as long-term interword relations of texts for more robust features $\hat{F}_v$ and $\hat{F}_t$. The triplet loss L_{triplet} and the proposed uni-modal entity loss L_{Entity} are computed to obtain the final loss L.

During the inference stage, for each image-text pair $(I, T) \in D_{\text{test}}$ of the test set, we use the MGRM-EL model to obtain the single-modal representation $\hat{F}_v$ and $\hat{F}_t$. It should be noted that during the inference process, the mask ratios τ_v and τ_t of the image and text are both set to 0 (meaning no masking). Then the cosine function $\cos(\cdot, \cdot)$ is used to calculate the cross-modal similarity distance S_{cal} between query and targets, and all test datasets are combined to generate a similarity matrix S. Finally, using the pairwise similarity in S, we generate the reranking of the top-10 retrieval results.

IV. Experiments

In this section, we report the experimental results on two popular benchmark datasets: RSICD [18] and RSITMD [9]. First, we introduce the implementation details, datasets, and evaluation metrics used in our experiments. Then we compare our approach against several state-of-the-art methods. Finally, we conduct an ablation study to demonstrate the contribution of individual modules in our approach, including mask-guided attention mechanism, relation modeling, uni-modal entity loss, and some other parameters.

A. Implementation Details

1) Experimental Settings: The proposed MGRM-EL is implemented in the Pytorch framework [59] on a single NVIDIA GTX 3090 GPU and trained in an end-to-end manner. During training, we resize the input image to 278 × 278. Random rotation and random crop to 256 are used as data augmentations. Adam optimizer [60] is adopted for training, where $\beta_1 = 0.9$, $\beta_2 = 0.999$, and the weight decay is default. The initial learning rate is $1e^{-4}$ and decays by 0.7 every 20 epochs. The batch size is set to 200 and the total number of epochs for training is 70 in the RSICD dataset and 50 in the RSITMD dataset according to their scale. To make the experiment more convincing, for each set of experiments, we conduct the same training and verification three times and take the average result. During inference, input images are resized to 256 × 256, and augmented only by normalization. Our Transformer framework [23] has four attention heads, and we use a three-layer Transformer encoder with a residual layer around each embedding update and layer norm, i.e., $x = 3$. The feature embedding dimension of both images and text is set to 512. The margin α is set to 0.2 for the triplet loss calculation. The domain radius of the DBSCAN [56] clustering algorithm is set to 0.15.

B. Dataset and Evaluation Metrics

1) Dataset: Two commonly used RS text-image datasets are used in our experiment. RSICD [18] has 10 921 samples with the image size of 224 × 224, which is the largest RS

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Algorithm 1 Training Procedure of the Proposed MGRM-EL

Require:

RS image-text train dataset $D_{train} = \{(I_1, T_1), (I_2, T_2), \ldots\}$ (I and T is one image and the corresponding sentence, respectively),
Image processing backbone $\mathcal{B}$,
Text processing backbone $\mathcal{G}$,
Random sampling module R_{mask},
Image mask ratio τ_v (the ratio of removed regions),
Text mask ratio τ_t (the ratio of removed words),
Relational modeling module MSA.

Repeat until convergence:

for each image-text pair $(I, T) \in D_{train}$ **do**
  Visual feature extraction:
  $F_v = \mathcal{B}(I; \theta_v)$
  $F_v' = R_{mask}(F_v; \tau_v)$
  $\hat{F}_v = \mathrm{MSA}_{r\to r}((v_i', \{v_1', \ldots, v_n'\}); (W_r^Q, W_r^K, W_r^V))$
  Textual feature extraction:
  $F_t = \mathcal{G}(T; \theta_t)$
  $F_t' = R_{mask}(F_t; \tau_t)$
  $\hat{F}_t = \mathrm{MSA}_{w\to w}((t_i', \{t_1', \ldots, t_m'\}); (W_w^Q, W_w^K, W_w^V))$
  Loss calculation:
  $P_{visual} = \mathrm{softmax}(\mathbb{W}_v \hat{F}_v + b_1)$
  $P_{textual} = \mathrm{softmax}(\mathbb{W}_t \hat{F}_t + b_2)$
  Calculate $L_{triplet}$ in (7),
  Calculate L_{Entity} in (13)
  $L = L_{triplet} + \lambda L_{Entity}$
  Update θ_v, θ_t, W_r^Q, W_r^K, W_r^V, W_w^Q, W_w^K, W_w^V, $\mathbb{W}_v$, $\mathbb{W}_t$ **by** L
end for
return $\mathcal{B}$, $\mathcal{G}$, $\mathrm{MSA}_{r\to r}()$, $\mathrm{MSA}_{w\to w}()$

Algorithm 2 Inference Procedure of the Proposed MGRM-EL

Require:

RS image-text test dataset $D_{test} = \{\{I_1, T_1\}, \{I_2, T_2\}, \ldots\}$ (I and T is one image and the corresponding sentence, respectively),
The proposed MGRM-EL model,
Mask ratio $\tau_v = 0$, $\tau_t = 0$,
Initialize the similarity matrix S,
Reranking module RR.

Calculate image-text similarity matrix:

for each image-text pair $(I, T) \in D_{test}$ **do**
  Visual feature extraction:
  $F_v = \mathcal{B}(I; \theta_v)$
  $F_v' = R_{mask}(F_v; \tau_v = 0)$
  $\hat{F}_v = \mathrm{MSA}_{r\to r}((v_i', \{v_1', \ldots, v_n'\}); (W_r^Q, W_r^K, W_r^V))$
  Textual feature extraction:
  $F_t = \mathcal{G}(T; \theta_t)$
  $F_t' = R_{mask}(F_t; \tau_t = 0)$
  $\hat{F}_t = \mathrm{MSA}_{w\to w}((t_i', \{t_1', \ldots, t_m'\}); (W_w^Q, W_w^K, W_w^V))$
  Similarity calculation:
  $S_{cal} = \cos(\hat{F}_v, \hat{F}_t)$
  Append S_{cal} to S
end for
return S
Generate top-10 reranking results:
  Target = RR(S)
return Target

text-image dataset in sample numbers. RSITMD [9] is a fine-grained dataset, in which some of the images are selected from the RSICD dataset, while others are from Google Earth. It has 23 715 captions for 4743 images and the image size is 256 × 256. Each RS image in RSICD and RSITMD has five corresponding sentences. In experiments, we follow the data partitioning approach in [9] and 80%, 10%, and 10% of the dataset are used as the training set, validation set, and test set, respectively.

2) Evaluation Metrics of Retrieval: Two evaluation indicators $R@K$ ($K = 1$, 5, and 10) and mR [61] are used to evaluate our method. The metric $R@K$ represents the ratio of ground truth appearing in the top K results, while mR is obtained with the average of all six recall rates of $R@K$ to evaluate the overall performance of the method. Besides, we add two metrics, $m_s R$ and $m_i R$, to evaluate the average of $R@K$ ($K = 1$, 5, and 10) on sentence and image retrieval, respectively.

3) Evaluation Metrics of Semantic Localization (SeLo): SeLo is an emerging task based on cross-modal retrieval, and it refers to the task of obtaining the most relevant locations in large-scale RS images using semantic text information. To demonstrate the effectiveness of the cross-modal semantic alignment, we visualize the SeLo map in the ablation studies and apply the common metrics R_{su}, R_{as}, R_{da}, and R_{mi} [13] in SeLo as evaluation metrics. Among them, R_{su} denotes the proportion of attention in GT areas, R_{as} indicates the shift between the attention generated by the model and the center of the GT region, and R_{da} reflects the discrete degree of attention in GT areas to verify the stability of the model in retrieval. The mean indicator R_{mi} comprehensively measures various aspects of SeLo, which provides a multivariate and reasonable quantitative score.

C. Comparison With State-of-the-Art Methods

We compare the performance of our proposed MGRM-EL with several state-of-the-art methods on RSICD and RSITMD datasets in Table I. The introduction of each model is as follows.

1) *VSE++ [62]:* VSE++ extracts image and text features by CNN and GRU and directly uses triplet loss to optimize them.
2) *SCAN [63]:* SCAN is based on VSE++, which utilizes Faster-RCNN [64] to extract image features and tries to align the target in the image with the target in the text.
3) *CAMP [65]:* CAMP adaptively controls the information flow for message passing across modalities. CAMP-triplet uses MLP on the fused features to predict the matching score, and adopts ranking loss for training supervision. CAMP-bce denotes training with only the hardest negative binary cross-entropy loss.

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4) *MTFN [66]:* MTFN uses a multimodal fusion network based on the idea of rank decomposition to calculate the distance of the embedding features.
5) *LW-MCR [14]:* LW-MCR utilizes knowledge distillation and contrast learning methods for lightweight retrieval models and sets up a multilevel and multimodal distillation loss. LW-MCR-b denotes the method only performs triplet loss optimization, and LW-MCR-d optimizes the LW-MCR-b method by using AMFMN as a teacher network.
6) *AMFMN [9]:* AMFMN can dynamically filter redundant features and is a multisource retrieval method with dynamic variable margin triplet loss. The difference between AMFMN-soft and AMFMN-fusion is the visual-guided attention (VGA) module, which is motivated by soft attention [67] and fusion attention [68], respectively.
7) *GaLR [10]:* GaLR fully adopts a MIDF module to efficiently integrate features of different levels and uses triplet loss to obtain model loss. Multivariate reranking (MR) is a ranking algorithm during inference.
8) *MGRM (Ours):* MGRM is our proposed method without proposed uni-modal entity loss.
9) *MGRM-EL w/o Cluster (Ours):* MGRM-EL w/o cluster is our proposed model with proposed loss but without clustering the text sentences.
10) *MGRM-EL (Ours):* MGRM-EL is our proposed complete model.

1) Quantitative Results: We compare our method with recent state-of-the-art RSCTIR methods on the RSICD and RSITMD datasets in Table I. The results show that the proposed MGRM-EL model achieves the best performance in terms of most metrics. For example, on the RSICD dataset, our model achieves the best performance in terms of all metrics on both sentence and image retrieval. For the model MGRM without uni-modal entity loss, the mR score reached 19.40. After adding the proposed uni-modal entity loss, the mR score of our MGRM-EL is increased to 21.42, 2.46 more than the best GaLR model proposed last year. The performance of MGRM-EL on RSICD, a large RS I2T dataset, strongly illustrates the robustness and effectiveness of the proposed method. On the RSITMD dataset, a few indicators (such as $R@1$ of sentence retrieval and image retrieval) of our model are not optimal. The reason is that the RSITMD dataset is relatively small. It can be seen that our method is still very competitive in terms of the top-5 ($R@5$), top-10 ($R@10$), $m_s R$, and $m_i R$ metrics, and the overall performance index mR has achieved state-of-the-art compared with the other methods.

2) Qualitative Results: Fig. 6 shows the image-text and text-image retrieval results on the RSICD dataset with the comparison of AMFMN, GaLR, and the proposed MGRM-EL. The AMFMN model directly uses global features for retrieval, resulting in many fine-grained regions that cannot be correctly identified. For example, AMFMN is easily confused with "basketball fields," "baseball fields" and "tennis courts" in the image-text retrieval task. GaLR considers both global features and local features, but does not take into account the spatial correlation of local areas, which confuses the spatial correlation of "green trees" and "playground." In contrast, the proposed MGRM-EL gets better retrieval results on similar sentences or images. We attribute this to the proposed uni-modal feature learning on entities and relations based on mask-guided relation modeling and entity loss.

D. Ablation Studies

In this section, ablation studies are given to systematically analyze the proposed MGRM-EL method on the RSICD dataset. Ten experimental settings shown in Table II are designed to analyze the influence of each module on retrieval effectiveness.

1) $\mathcal{T}_0$: Use only visual features extracted by CNN (ResNet18) and textual features extracted by Seq2Vec (Skipthought).
2) $\mathcal{T}_1$: Add visual Transformer based on $\mathcal{T}_0$ to model intraimage relations.
3) $\mathcal{T}_2$: Add textual Transformer based on $\mathcal{T}_0$ to model intratext relations.
4) $\mathcal{T}_3$: Merge $\mathcal{T}_1$ and $\mathcal{T}_2$, i.e., simultaneously add visual and textual Transformer based on $\mathcal{T}_0$ to model the visual and textual relations for each modality.
5) $\mathcal{M}_1$: Add mask-guided strategy based on $\mathcal{T}_3$ to learn the visual attention.
6) $\mathcal{M}_2$: Add mask-guided strategy based on $\mathcal{T}_3$ to learn the textual attention.
7) $\mathcal{M}_3$: Merge $\mathcal{M}_1$ and $\mathcal{M}_2$, i.e., simultaneously add mask-guided strategy to learn the visual and textual attention for each modality.
8) $\mathcal{U}_1$: Add visual entity loss based on $\mathcal{M}_3$ to learn the single-modal visual feature.
9) $\mathcal{U}_2$: Add textual entity loss based on $\mathcal{M}_3$ to learn the single-modal textual feature.
10) $\mathcal{U}_3$: Merge $\mathcal{U}_1$ and $\mathcal{U}_2$, i.e., the complete model with Backbone, Transformer, Mask strategy, and uni-modal entity loss.

The experimental results are shown in Table II, and we can roughly divide the ablation experiments into three groups.

1) Group ($\mathcal{T}_0$, $\mathcal{T}_1$, $\mathcal{T}_2$, $\mathcal{T}_3$) is to explore the impact of Transformer relation modeling.
2) Group ($\mathcal{T}_3$, $\mathcal{M}_1$, $\mathcal{M}_2$, $\mathcal{M}_3$) is to explore the impact of mask strategy on model performance.
3) Group ($\mathcal{M}_3$, $\mathcal{U}_1$, $\mathcal{U}_2$, $\mathcal{U}_3$) is to explore the impact of the proposed uni-modal entity loss.

From this, we can draw the following conclusions. Compared with $\mathcal{T}_0$, after adding the Transformer of visual features to the $\mathcal{T}_1$ model, mR increased by 0.43. After the $\mathcal{T}_2$ model is added to the Transformer with textual features, the mR has increased by 5.98, especially the $R@1$, $R@5$, and $R@10$ of Image Retrieval have increased by 2.45, 11.87, and 19.75, respectively, which demonstrated that a more superior feature is conducted after adding the Transformer and proved insufficient of Seq2Vec textual backbone (Skipthought). After adding the Transformer of visual features and text features to the $\mathcal{T}_3$ model, both Sentence Retrieval and Image Retrieval have been steadily improved, and mR has increased by 6.88 compared with $\mathcal{T}_0$, reaching 17.96.

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TABLE I

COMPARISON OF RETRIEVAL PERFORMANCE ON RSICD AND RSITMD TESTSETS. THE **FIRST** AND SECOND HIGHEST SCORES ARE REPRESENTED BY BOLD AND UNDERLINED TEXT, RESPECTIVELY

Method	RSICD dataset									RSITMD dataset								
	Sentence Retrieval				Image Retrieval				mR	Sentence Retrieval				Image Retrieval				mR
	R@1	R@5	R@10	m_sR	R@1	R@5	R@10	m_iR		R@1	R@5	R@10	m_sR	R@1	R@5	R@10	m_iR	
VSE++	3.38	9.51	17.46	10.12	2.82	11.32	18.10	10.75	10.43	10.38	27.65	39.60	25.88	7.79	24.87	38.67	23.78	24.83
SCAN t2i	4.39	10.90	17.64	10.98	3.91	16.20	26.49	15.53	13.25	10.18	28.53	38.49	25.73	10.10	28.98	43.53	27.54	26.64
SCAN i2t	5.85	12.89	19.84	12.86	3.71	16.40	26.73	15.61	14.23	11.06	25.88	39.38	25.44	9.82	29.38	42.12	27.11	26.28
CAMP-triplet	5.12	12.89	21.12	13.04	4.15	15.23	27.81	15.73	14.39	11.73	26.99	38.05	25.59	8.27	27.79	44.34	26.80	26.20
CAMP-bce	4.20	10.24	15.45	9.96	2.72	12.76	22.89	12.79	11.38	9.07	23.01	33.19	21.76	5.22	23.32	38.36	22.30	22.03
MTFN	5.02	12.52	19.74	12.43	4.90	17.17	29.49	17.19	14.81	10.40	27.65	36.28	24.78	9.96	31.37	45.84	29.06	26.92
LW-MCR(b)	4.57	13.71	20.11	12.80	4.02	16.47	28.23	16.24	14.52	9.07	22.79	38.05	23.30	6.11	27.74	49.56	27.80	25.55
LW-MCR(d)	3.29	12.52	19.93	11.91	4.66	17.51	30.02	17.40	14.66	10.18	28.98	39.82	26.33	7.79	30.18	49.78	29.25	27.79
AMFMN-soft	5.05	14.53	21.57	13.72	5.05	19.74	31.04	18.61	16.02	11.06	25.88	39.82	25.59	9.82	33.94	51.90	31.89	28.74
AMFMN-fusion	5.39	15.08	23.40	14.62	4.90	18.28	31.44	18.21	16.42	11.06	29.20	38.72	26.33	9.96	34.03	52.96	32.32	29.32
AMFMN-sim	5.21	14.72	21.57	13.83	4.08	17.00	30.60	17.23	15.53	10.63	24.78	41.81	25.74	**11.51**	34.69	54.87	33.69	29.72
GaLR w/o MR	6.50	18.91	29.70	18.37	5.11	19.57	31.92	18.87	18.62	13.05	30.09	42.70	28.61	10.47	36.34	53.35	33.39	31.00
GaLR with MR	6.59	19.85	31.04	19.16	4.69	19.48	32.13	18.77	18.96	**14.82**	31.64	42.48	29.65	11.15	36.68	51.68	33.17	31.41
MGRM(ours)	7.29	20.28	31.59	19.72	5.68	19.77	31.79	19.08	19.40	12.17	30.75	44.91	29.28	9.91	36.81	55.93	34.22	31.75
MGRM-EL w/o cluster(ours)	**7.96**	20.77	31.38	20.04	**6.15**	**21.57**	34.86	**20.86**	20.45	13.39	31.34	45.42	30.05	11.01	36.74	55.81	34.52	32.28
MGRM-EL (ours)	7.41	**23.24**	**35.32**	**21.99**	5.75	21.23	**35.55**	20.84	**21.42**	13.51	**31.87**	**46.27**	**30.55**	11.11	**37.22**	**56.61**	**34.98**	**32.76**

TABLE II

ABLATION EXPERIMENTS OF DIFFERENT COMPOSITIONS ON THE RSICD TESTSET

Abalation Model	Backbone		Transformer		Mask		Entity Loss		Sentence Retrieval			Image Retrieval			mR
	CNN	Seq2Vec	Visual	Textual	Visual	Textual	Visual	Textual	R@1	R@5	R@10	R@1	R@5	R@10	
$\mathcal{T}_0$	✓	✓							4.48	15.37	26.17	1.87	7.06	11.53	11.08
$\mathcal{T}_1$	✓	✓	✓						4.76	15.37	26.90	1.96	7.58	12.50	11.51
$\mathcal{T}_2$	✓	✓		✓					4.31	16.38	27.17	4.32	18.93	31.28	17.06
$\mathcal{T}_3$	✓	✓	✓	✓					4.28	18.00	32.30	4.99	18.48	29.73	17.96
$\mathcal{M}_1$	✓	✓	✓	✓	✓				4.23	17.85	32.53	5.12	18.76	31.34	18.30
$\mathcal{M}_2$	✓	✓	✓	✓		✓			5.28	18.24	32.58	4.20	18.23	30.31	18.14
$\mathcal{M}_3$	✓	✓	✓	✓	✓	✓			7.29	20.28	31.59	5.68	19.77	31.79	19.40
$\mathcal{U}_1$	✓	✓	✓	✓	✓	✓	✓		7.04	19.76	30.47	**5.89**	20.51	33.19	19.48
$\mathcal{U}_2$	✓	✓	✓	✓	✓	✓		✓	**7.87**	18.76	30.28	5.56	20.71	34.71	19.65
$\mathcal{U}_3$	✓	✓	✓	✓	✓	✓	✓	✓	7.41	**23.24**	**35.32**	5.75	**21.23**	**35.55**	**21.42**

Compared with $\mathcal{T}_3$, after adding visual mask-attention to $\mathcal{M}_1$, Image Retrieval has significantly improved compared to Sentence Retrieval, and $R@1$, $R@5$, and $R@10$ have increased by 0.13, 0.28, and 1.61, respectively. After $\mathcal{M}_2$ added textual mask-attention, Sentence Retrieval has improved compared with Image Retrieval, and $R@1$, $R@5$, and $R@10$ have increased by 1.00, 0.24, and 0.28, respectively. The $\mathcal{M}_3$ has an overall impressive improvement over the $\mathcal{T}_3$, with mR up 1.44 and reaching the new SoTA of 19.40. It demonstrates that the proposed mask strategy improves the robustness of the model.

Compared with $\mathcal{M}_3$, $\mathcal{U}_1$ has added the entity loss of visual features, which has achieved a good improvement in Image Retrieval. $R@1$, $R@5$, and $R@10$ have improved by 0.21, 0.74, and 1.40, respectively, and almost in Sentence Retrieval has no effect. After $\mathcal{U}_2$ added the entity loss of text features, mR increased by 0.25. It can be proved that the proposed uni-modal entity loss is effective for image-text retrieval. The mR of the complete model $\mathcal{T}_3$ reaches 21.42, which is 2.46 higher than the current highest score GaLR.

E. Performance Analysis of Relation Modeling

In Table II, the experimental group ($\mathcal{T}_0$, $\mathcal{T}_1$, $\mathcal{T}_2$, $\mathcal{T}_3$) shows the impact of the proposed Transformer relations modeling framework on model performance. Furthermore, we analyze the effect of the proposed framework based on the Transformer by cosine similarity. Here, a cosine similarity ($\cos\langle\cdot,\cdot\rangle$) is used to measure the relations between different words. The $\cos\langle\cdot,\cdot\rangle$ value ranges from 0 (weak correlation) to 1 (strong correlation). This means that the larger the value, the stronger the relations between two different words in the sentence.

As shown in Fig. 7, we compute the cosine similarity between all words for the query "a cyan ship with cargo on the black sea is unloading at the port" based on Seq2Vec Backbone and MGRM-EL. Here, (a) Seq2Vec Backbone represents the features obtained only using the text feature extractor Skipthought [45]; (b) MGRM-EL represents the features obtained after semantic information modeling based on Transformer. It can be seen that the proposed MGRM-EL method can effectively be aware of the correlation among different words. For instance, it is shown that there are improved correlations between some words, $\cos\langle ship,\ cargo\rangle$ improved from 0.62120 to 0.79471, which is an increase of 0.17351; $\cos\langle cargo,\ port\rangle$ improved from 0.44851 to 0.72738, an increase of 0.27887.

Table III presents the analysis results regarding the effects of the Seq2Vec Backbone and MGRM-EL. We selected six common sets of words, including four sets of positive samples and two sets of negative samples. It can be observed the proposed MGRM-EL can make the relations between words more accurate. For example, the relation between "port" and "ship" becomes stronger, and the relation between "a" and "the" gets weaker.

F. Performance Analysis of Mask-Guided Attention

In this section, we quantitatively and qualitatively analyze the impact of mask-guided attention by SeLo. SeLo [13] is to locate objects in large-scale RS images using semantic text

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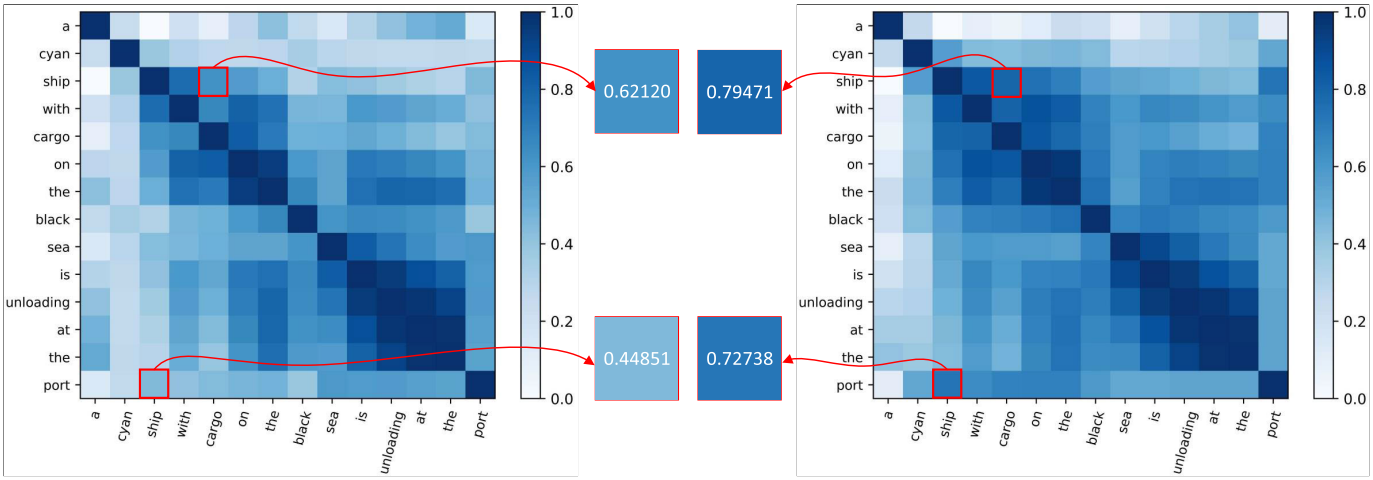


Fig. 7. Illustration of cosine similarity of text word features in backbone and MGRM-EL with the query of "a cyan ship with cargo on the black sea is unloading at the port." The left is obtained from Seq2Vec backbone, and the right is from MGRM-EL (ours). As shown in the figure example, the cosine similarity between "ship" and "cargo" and the cosine similarity between "ship" and "port" has been improved. Best viewed in color.

TABLE III

COSINE SIMILARITY ANALYSIS OF WORD FEATURES IN THE TEXT. THE OPERATION $\cos\langle\cdot,\cdot\rangle$ IS SHORT FOR A COSINE SIMILARITY

	Seq2Vec Backbone	MGRM-EL	Growth
$\cos\langle a,\ the\rangle$	0.41444	0.22644	↓ 0.18800
$\cos\langle ship,\ cargo\rangle$	0.62120	0.79471	↑ 0.17351
$\cos\langle sea,\ ship\rangle$	0.43303	0.53906	↑ 0.10603
$\cos\langle port,\ ship\rangle$	0.44851	0.72738	↑ 0.27887
$\cos\langle a,\ port\rangle$	0.13642	0.10286	↓ 0.03356
$\cos\langle cargo,\ port\rangle$	0.44184	0.67774	↑ 0.23590

TABLE IV

QUANTITATIVE COMPARISON OF SELO OF AMFMN, MGRM-EL W/O MASK, AND MGRM-EL WITH EIGHT QUERY EXAMPLES. THE HIGHEST SCORE HAS BEEN BOLDED

	$\uparrow R_{su}$	$\uparrow R_{da}$	$\downarrow R_{as}$	$\uparrow R_{mi}$
Sentence 1	Several green baseball fields next to each other.			
AMFMN	0.7165	0.4093	0.2055	0.6670
MGRM-EL w/o Mask	0.9343	**1.0000**	0.1350	0.9265
MGRM-EL	**0.9784**	**1.0000**	**0.1136**	**0.9516**
Sentence 2	A white airplane ready to take off on a grayblack runway.			
AMFMN	0.4078	0.0000	1.0000	0.1631
MGRM-EL w/o Mask	0.7657	0.3971	0.3117	0.6465
MGRM-EL	**0.8706**	**0.4536**	**0.1343**	**0.7646**
Sentence 3	A white plane parked in a tawny clearing inside the airport.			
AMFMN	0.9166	**1.0000**	0.2642	0.8742
MGRM-EL w/o Mask	0.9897	**1.0000**	0.1874	0.9303
MGRM-EL	**0.9923**	**1.0000**	**0.1262**	**0.9527**
Sentence 4	Ships without cargo floating on the black sea are docked in the port.			
AMFMN	0.5323	0.5000	0.2167	0.6121
MGRM-EL w/o Mask	0.5742	**1.0000**	0.5045	0.6531
MGRM-EL	**0.5770**	0.6584	**0.1679**	**0.6867**
Sentence 5	A green baseball field is surrounded by bare land.			
AMFMN	0.5984	**1.0000**	0.1946	0.7712
MGRM-EL w/o Mask	0.7409	**1.0000**	**0.0768**	0.8695
MGRM-EL	**0.9786**	**1.0000**	0.1152	**0.9511**
Sentence 6	The green football field is surrounded by a red track.			
AMFMN	0.8297	0.7659	0.0765	0.8466
MGRM-EL w/o Mask	**0.9058**	**1.0000**	0.2610	0.8709
MGRM-EL	0.8418	**1.0000**	**0.0621**	**0.9150**
Sentence 7	The green baseball field is next to a playground with a black track.			
AMFMN	0.7072	0.3333	0.5930	0.5087
MGRM-EL w/o Mask	0.5892	**0.6667**	**0.3098**	0.6439
MGRM-EL	**0.8295**	**0.6667**	0.3919	**0.7113**
Sentence 8	Three small green lakes surrounded by dense green forest.			
AMFMN	0.5813	0.2130	0.4361	0.4831
MGRM-EL w/o Mask	0.6060	0.3613	0.2189	0.6061
MGRM-EL	**0.6909**	**0.3669**	**0.2099**	**0.6446**

information. It can demonstrate the location accuracy of the image-text retrieval model.

1) Quantitative Analysis: In Table II, the experiment group ($\mathcal{T}_3$, $\mathcal{M}_1$, $\mathcal{M}_2$, $\mathcal{M}_3$) shows the effect of mask-guided attention on model performance. Moreover, we illustrate the impact of the masking strategy more intuitively through the performance of SeLo. Table IV shows the quantitative comparison of SeLo of AMFMN, MGRM-EL w/o Mask (i.e., MGRM-EL with Transformer but without mask-guided strategy), and MGRM-EL. To fully demonstrate the experimental results, we cite eight sentence queries. It can be seen that compared with AMFMN [9], the proposed MGRM-EL method has an impressive improvement in various indicators. After adding mask-guided attention, R_{su} that represents the proportion of attention in GT areas and mean indicator R_{mi} have been significantly improved. For example, in Sentence 2, the performance of R_{su}, R_{da}, R_{as}, and R_{mi} increased by 10.49%, 5.65%, 17.74%, and 11.81%, respectively. In Sentence 5, the performance of R_{su} and R_{mi} increased by 23.77% and 8.16%, respectively. This shows that after adding the masking strategy, the model has learned a more robust feature-matching relationship, which is very effective.

2) Qualitative Analysis: In order to better show the results of the experiment, we apply the SeLo task to obtain the visualized results. Following the work in [13], the scene image is cut into nonoverlapping slices with different scales and the similarity scores of all slices in the same scale form one similarity probability distribution map. We take the pixel-level average of the regional similarity on different scales to form heatmaps. The located image is obtained by simply adding the heatmaps to the original input image and increasing the transparency of the heatmaps. We visualize the SeLo of eight

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TABLE V
QUANTITATIVE COMPARISON OF SELO OF MGRM (MGRM-EL WITHOUT THE PROPOSED ENTITY LOSS) AND MGRM-EL. THE HIGHEST SCORE HAS BEEN BOLDED

	$\uparrow R_{su}$	$\uparrow R_{da}$	$\downarrow R_{as}$	$\uparrow R_{mi}$
Sentence 1	There is a green pond next to the gray road.			
MGRM	0.6652	0.3899	0.1491	0.6613
MGRM-EL	**0.8477**	**1.0000**	**0.0144**	**0.9341**
Sentence 2	Three big green soccer fields next to each other.			
MGRM	**0.9570**	0.5000	0.2729	0.7623
MGRM-EL	0.8979	**1.0000**	**0.1733**	**0.8985**

sentence examples in images, where the ground-truth values are surrounded by red boxes, as shown in Fig. 8. Each group is located image and heatmap from left to right. It can be seen that compared with AMFMN, the proposed MGRM-EL has a more accurate SeLo. After adding the masking strategy, the accuracy and robustness of SeLo are further enhanced.

G. Performance Analysis of Uni-Modal Entity Loss

In Table II, the experimental group ($\mathcal{M}_3, \mathcal{U}_1, \mathcal{U}_2, \mathcal{U}_3$) shows the influence of the proposed uni-modal entity loss on model performance. In this section, we analyze the impact of the proposed Entity loss by SeLo.

In Fig. 9, we show the SeLo of MGRM (MGRM-EL without the proposed Entity Loss) and the complete model MGRM-EL under two examples. It can be seen that MGRM-EL has a more accurate SeLo than MGRM. To more intuitively reflect the superiority of Entity Loss, we quantify SeLo through evaluation indicators, as shown in Table V. In Sentence 1, MGRM-EL was 18.25%, 61.01%, 13.47%, and 27.28% higher than MGRM in R_{su}, R_{da}, R_{as}, and R_{mi}, respectively; In Sentence 2, R_{mi} increased by 13.62%. It can be demonstrated that after adding Entity Loss, the model has learned more robust and accurate image-text matching relations.

H. Settings Analysis of MGRM-EL

In this section, we conduct ablation studies to quantitatively analyze the effect of different settings for mask ratio, the Transformer encoder layers, the cluster of entity loss, and the weight of loss.

1) Effect on Mask Ratio: In this ablation study, we conduct experiments with different ratios of masking ratios. According to the analysis of the characteristics of RS images, we set the default image mask ratio to 50% and increase the text mask ratio from 0% to 75%. The values of the corresponding mR are 18.30 (0%), 21.42 (25%), 19.98 (50%), and 18.05 (75%), respectively. It can be seen that the best ratio of the text mask is 25%. At the same time, we set the default text mask ratio to 25% and increased the image mask ratio from 0% to 75%. We obtained mR is 18.14 (0%), 20.25 (25%), 21.42 (50%), and 19.47 (75%), respectively. It can be seen that the best ratio of the image mask is 50%. So in the above experiment, we set the mask ratio of the image and text to 25% and 50%, respectively.

2) Effect on Transformer Encoder Layers: The performance that MGRM-EL achieved can be affected by the layers of multihead attention and the number of encoder layers in the Transformer, which can model correlations among features of different regions and words. We conduct experiments by changing the number of multihead attention to evaluate the importance of multiheaded self-attention and observe that the performance improves as the multihead attention layers increase. Given the limitation of resources, we set the number of multihead attention to 4 in our experiments. In addition, we conduct experiments on the effect of the number of encoder layers. As shown in Fig. 10(a), when the number of layers is set to 2, 3, and 4, the mR is 19.54, 21.42, and 19.93, respectively. So we set the number of encoder layers to 3 in our experiments.

3) Effect on the Cluster of Entity Loss: Considering that there will be many similarities or repetitions between texts in the dataset, we use the DBSCAN [56] as the clustering algorithm to generate pseudo labels to calculate the cross-entropy loss of textual features. It is a density-based clustering algorithm that groups data points based on their density in the feature space. It requires two parameters: ε (the maximum distance between points to be considered in the same neighborhood) and minPts (the minimum number of points required to form a dense region). In our method, we set minPts to 1 and performed ablation experiments on the setting of ε. As shown in Fig. 10(b), when the ε of the cluster is set to 0.10, 0.15, and 0.20, the mR is 19.65, 21.42, and 20.70, respectively. It can be seen that when ε is set to 0.15, the model performs the best.

4) Effect on the Weight of Loss: In our proposed model, two loss functions are involved: L_{triplet} and L_{Entity}, as shown in (14). The parameter λ represents the trade-off of L_{Entity}. We evaluate the effect of weight parameter λ in (14) as shown in Fig. 10(c). When $\lambda = 0.0$, L_{Entity} is discarded and can not discriminate unimodal entity and mR is 19.40. With the injection of L_{Entity}, mR improves significantly. When lambda is set to 0.25, 0.5, 0.75, 1.00, and 1.25, respectively, its mR is 19.58, 20.32, 20.88, 21.42, and 19.77, respectively. MGRM-EL model achieves the best performance when λ is 1.0, where the well-learned characteristic of the uni-modal entity is.

I. Analysis of Time Consumption

In this section, we compare the retrieval time consumption of the proposed method with two other state-of-the-art RSCTIR methods, AMFMN [9] and GaLR [10]. Following the work in [9] and [10], two reasonable evaluation indicators, inference time (IT) and evaluation time (EvaT), are applied to evaluate the retrieval performance of computation speed. The IT metric denotes the average computation time of one retrieval on the test dataset spent by feature extraction and cross-modal similarity calculation, while EvaT refers to the total time of all retrievals spent in the test dataset. We also provide the number of parameters utilized during model inference and they are measured in millions (M). We conducted multiple experiments on the two public datasets, RSICD and

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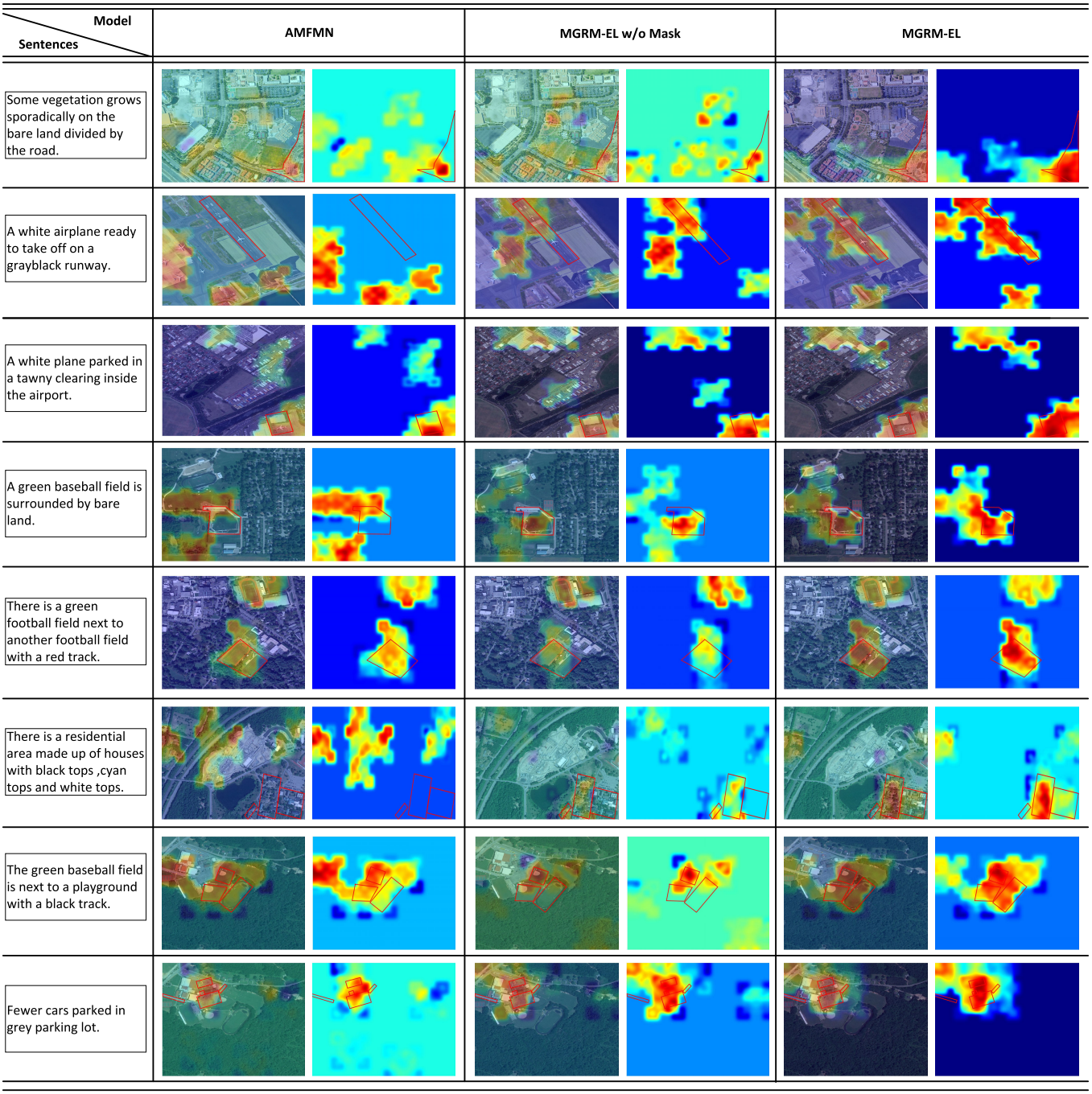


Fig. 8. Visualized qualitative comparison of SeLo of AMFMN, MGRM-EL w/o mask, and MGRM-EL with eight query examples. The area enclosed by the red box is the ground truth.

RSITMD, and all experiments are run on the same platform for fair comparison, which includes Intel[1] Xeon[1] CPU E5-2699C v4 at 2.20 GHz and a single NVIDIA GeForce RTX 3090 GPU.

Table VI shows the computation time and parameter comparison of different methods. Except for the proposed MGRM-EL with three-layer transformer encoders, we additionally introduce a lightweight version called MGRM-EL-s, denoting the model with a Transformer layer set to 2. The MGRM-EL-s reduces the number of parameters to 41.74M. In contrast to GaLR with 46.89 parameters, our MGRM-EL-s model achieves higher accuracy in mR and less time consumption in both IT and EvaT on both two datasets. Although AMFMN performs an advantage in terms of IT and EvaT, its retrieval accuracy falls the shortest performance in terms of mR. Overall, our proposed MGRM-EL model approaches GaLR in terms of computation time consumption and achieves superior accuracy performance. For applications that are limited in time, the lightweight MGRM-EL-s version could be an alternate for balancing accuracy and time.

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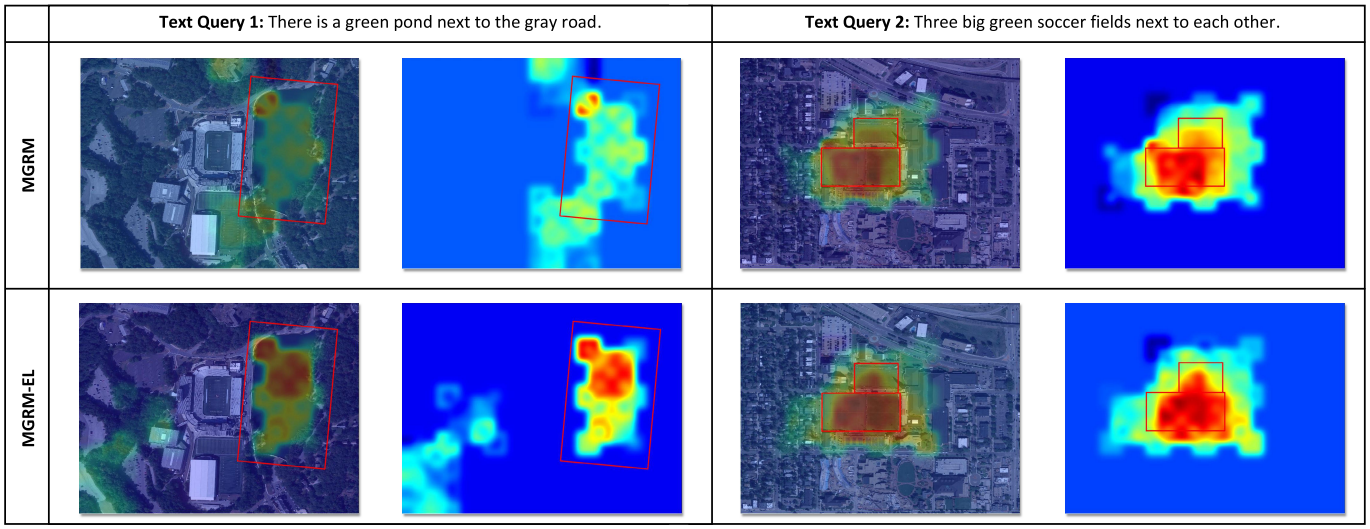


Fig. 9. Visualized comparison of SeLo of MGRM (MGRM-EL without the proposed entity loss) and MGRM-EL. The area enclosed by the red box is the ground truth.

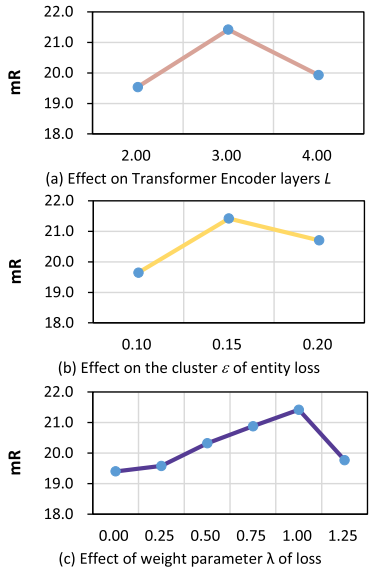


Fig. 10. Effect of parameter setting on the RSICD dataset. (a) Effect on transformer encoder layers L. (b) Effect on the cluster ε of entity loss. (c) Effect of weight parameter λ in (14).

TABLE VI
TIME-CONSUMING AND PARAMETERS (PARA.) COMPARISON OF DIFFERENT METHODS

Method	RSICD			RSITMD			Para.(M)
	IT(s)	EvaT(s)	mR	IT(s)	EvaT(s)	mR	
AMFMN	0.094	87.94	16.42	0.083	19.30	29.72	35.94
GaLR	0.107	96.56	18.96	0.117	26.91	31.41	46.89
MGRM-EL-s	0.106	93.97	19.08	0.110	22.24	31.69	41.74
MGRM-EL	0.125	102.39	21.42	0.129	25.25	32.76	73.35

V. CONCLUSION

In this article, an RS cross-modal retrieval method named MGRM-EL is proposed. We propose a mask-guided attention strategy for the redundant information of RS images and text. The Transformer-based relations modeling framework is proposed to model the relationship of intraregions and intrawords. Specifically, a novel uni-modal entity loss is proposed to alleviate the weakness of triplet loss application in RS cross-modal retrieval tasks. Extensive experiments and analysis on multiple RS text-image datasets demonstrate the effectiveness and robustness of the proposed method for RS retrieval.

In future, as there are highly similar sub-classes in texts and images, we believe that exploiting the fine-grained learning task will improve the cross-modal semantic retrieve performance. In addition, a more lightweight model transplanted via precision quantization, model pruning and knowledge distillation approaches plays a vital role for the online satellite-borne or airborne applications.

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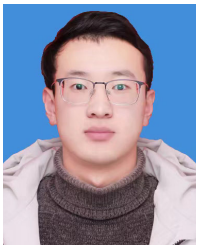
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